Chapter 2. Basic Properties of Fourier Series

Uniqueness, convolution, good kernels, Fejer and Poisson

Contents

2.1 Examples and formulation of the problem **2**
2.1.1 Main definitions and some examples 2

2.2 Uniqueness of Fourier series **3**

2.3 Convolutions **4**

2.4 Good kernels **5**

2.5 Cesàro and Abel summability: applications to Fourier series **6**
2.5.1 Cesàro means and summation 6
2.5.2 Fejér's theorem 6
2.5.3 Abel means and summation 7
2.5.4 The Poisson kernel and Dirichlet's problem in the unit disc 7

2.1 Examples and formulation of the problem

Motivation. We fix what "function" and "Fourier series" mean (Riemann integrable functions on the circle) and pose the basic question: in what sense does the Fourier series of f converge back to f?

2.1.1 Main definitions and some examples

A bounded function f on $[a,b]$ is (Riemann) **integrable** if $\sup_P L(P,f) = \inf_P U(P,f)$ over partitions P; continuous and piecewise continuous functions are integrable (see the Appendix). Complex $f = u + iv$ is integrable if u and v are. A function on the circle is a 2π-periodic function on $\mathbb{R}$, equivalently a function $F(e^{i\theta}) = f(\theta)$ on $\{|z| = 1\}$; we call such f integrable on the circle if it is integrable on one (hence every) period.

Definition (Fourier coefficients and series)

For f integrable on $[a,b]$, $L = b-a$, the n-th **Fourier coefficient** is

$$\hat{f}(n) = \frac{1}{L}\int_a^b f(x)e^{-2\pi inx/L}\,dx, \qquad n \in \mathbb{Z},$$

and the **Fourier series** is $f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n)e^{2\pi inx/L}$. On the circle ($L = 2\pi$): $\hat{f}(n) = a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)e^{-in\theta}\,d\theta$. The **$N$-th partial sum** is $S_N(f)(\theta) = \sum_{|n|\le N} \hat{f}(n)e^{in\theta}$. A **trigonometric polynomial** is $\sum_{|n|\le N} c_n e^{in\theta}$, of degree N if $c_{\pm N} \neq 0$.

Basic question. In what sense does $S_N(f) \to f$ as $N \to \infty$?

Example (Sawtooth)

For $f(\theta) = \frac{\pi-\theta}{2}$ on $0 < \theta < 2\pi$ (extended periodically, $f(0) = 0$): $\hat{f}(0) = 0$ and, integrating by parts, $\hat{f}(n) = \frac{1}{2in}$ for $n \neq 0$. So $f(\theta) \sim \sum_{n\neq 0} \frac{e^{in\theta}}{2in} = \sum_{n\ge 1} \frac{\sin n\theta}{n}$.

Example

For $f(\theta) = |\theta|$ on $[-\pi,\pi]$: $\hat{f}(0) = \frac{\pi}{2}$ and $\hat{f}(n) = \frac{-1+(-1)^n}{\pi n^2}$ for $n \neq 0$, so $f(\theta) \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{n \text{ odd}\ge 1} \frac{\cos n\theta}{n^2}$.

Example

For $f(\theta) = \theta$ on $(-\pi,\pi)$: $\hat{f}(0) = 0$, $\hat{f}(n) = \frac{(-1)^{n+1}}{in}$, so $f \sim 2\sum_{n\ge 1}(-1)^{n+1}\frac{\sin n\theta}{n}$.

Example (Dirichlet kernel)

$D_N(\theta) = \sum_{|n|\le N} e^{in\theta} = \frac{\sin((N+\frac{1}{2})\theta)}{\sin(\theta/2)}$; it is the trigonometric polynomial with $\hat{D}_N(n) = 1$ for $|n| \le N$ and 0 otherwise. *Proof:* with $\omega = e^{i\theta}$, $\sum_0^N \omega^n = \frac{1-\omega^{N+1}}{1-\omega}$ and $\sum_{-N}^{-1} \omega^n = \frac{\omega^{-N}-1}{1-\omega}$, so $D_N = \frac{\omega^{-N}-\omega^{N+1}}{1-\omega}$; multiply numerator and denominator by $\omega^{-1/2}$.

Example (Poisson kernel)

For $0 \le r < 1$, $P_r(\theta) = \sum_{n\in\mathbb{Z}} r^{|n|} e^{in\theta} = \frac{1-r^2}{1-2r\cos\theta+r^2}$ (absolutely convergent series; sum the two geometric series).

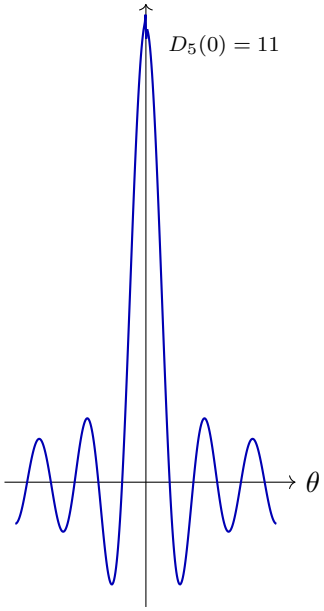


Figure 1: The Dirichlet kernel D_5: it takes negative values and $\int |D_N|$ grows like $\log N$.

2.2 Uniqueness of Fourier series

Motivation. If all Fourier coefficients of f vanish, then f vanishes at its points of continuity. Trigonometric polynomials that peak sharply at one point make this rigorous, and it already yields uniform convergence when $\sum |\hat{f}(n)| < \infty$, in particular for C^2 functions.

Theorem 2.1

Let f be integrable on the circle with $\hat{f}(n) = 0$ for all n. Then $f(\theta_0) = 0$ whenever f is continuous at θ_0.

Proof.

Step 1 (setup). First let f be real-valued. Translating (which multiplies $\hat{f}(n)$ by a unimodular factor), assume f is defined on $[-\pi,\pi]$, $\theta_0 = 0$, and suppose $f(0) > 0$. Since $\hat{f}(n) = 0$ for all n, $\int_{-\pi}^{\pi} f(\theta)p(\theta)\,d\theta = 0$ for every trigonometric polynomial p. We construct p_k with $\int f p_k \to \infty$.

Step 2 (the polynomials). Choose $0 < \delta \le \pi/2$ with $f(\theta) > f(0)/2$ for $|\theta| < \delta$. Let $p(\theta) = \epsilon + \cos\theta$ with $\epsilon > 0$ so small that $|p(\theta)| < 1 - \epsilon/2$ for $\delta \le |\theta| \le \pi$. Choose $0 < \eta < \delta$ with $p(\theta) \ge 1 + \epsilon/2$ for $|\theta| < \eta$. Let $p_k = p^k$, a trigonometric polynomial. Let B bound $|f|$.

Step 3 (estimates). On $\delta \le |\theta| \le \pi$: $\left|\int f p_k\right| \le 2\pi B(1-\epsilon/2)^k \to 0$. On $\eta \le |\theta| < \delta$: $p(\theta) \ge 0$ (as $\delta \le \pi/2$) and $f > 0$, so $\int f p_k \ge 0$ there. On $|\theta| < \eta$: $\int f p_k \ge 2\eta \frac{f(0)}{2}(1+\epsilon/2)^k \to \infty$. Hence $\int f p_k \to \infty$, contradicting $\int f p_k = 0$. So $f(0) \le 0$; applying this to $-f$, $f(0) = 0$.

Step 4 (complex f). Write $f = u + iv$ with $u = \frac{f+\bar{f}}{2}$, $v = \frac{f-\bar{f}}{2i}$. Since $\hat{\bar{f}}(n) = \overline{\hat{f}(-n)} = 0$, u and v have vanishing Fourier coefficients and are continuous at θ_0; apply the real case. ∎

Corollary 2.2

If f is continuous on the circle and $\hat{f}(n) = 0$ for all n, then $f = 0$.

Proof. Theorem 2.1 at every point. ■

Corollary 2.3

If f is continuous on the circle and $\sum_n |\hat{f}(n)| < \infty$, then $S_N(f) \to f$ uniformly: $\lim_{N\to\infty} S_N(f)(\theta) = f(\theta)$ uniformly in θ.

Proof. By the Weierstrass M-test, $S_N(f)$ converges uniformly to a continuous $g(\theta) = \sum \hat{f}(n)e^{in\theta}$. Uniform convergence allows term-by-term integration, so $\hat{g}(n) = \hat{f}(n)$. By Corollary 2.2 applied to $f - g$, $f = g$. ■

Corollary 2.4

If f is twice continuously differentiable on the circle, then $\hat{f}(n) = O(1/|n|^2)$, so the Fourier series of f converges absolutely and uniformly to f.

Proof. Integrating by parts twice (boundary terms cancel by periodicity), $2\pi\hat{f}(n) = \frac{1}{in}\int f'(\theta)e^{-in\theta}d\theta = \frac{-1}{n^2}\int f''(\theta)e^{-in\theta}d\theta$, so $|n|^2|\hat{f}(n)| \le \frac{1}{2\pi}\int |f''|$. Since $\sum 1/n^2 < \infty$, Corollary 2.3 applies. ■

Remark (on the hypotheses)

In fact $f \in C^1$ is enough (Exercise in the book), and if f is C^k then $\hat{f}(n) = o(1/|n|^k)$. In general, the smoother f, the faster $\hat{f}(n)$ decays.

2.3 Convolutions

Motivation. Partial sums and the various means of a Fourier series are all convolutions $f * K$ with a kernel. Convolution turns into multiplication of Fourier coefficients, and it smooths: $f * g$ is continuous even if f, g are only integrable.

Definition (Convolution)

For 2π-periodic integrable f, g on $\mathbb{R}$, $(f * g)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)g(x-y)\,dy$. For instance $S_N(f)(x) = \sum_{|n|\le N} \hat{f}(n)e^{inx} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\sum_{|n|\le N} e^{in(x-y)}dy = (f * D_N)(x)$.

Proposition 3.1

Let f, g, h be 2π-periodic integrable functions. Then:
(i) $f * (g + h) = (f * g) + (f * h)$.
(ii) $(cf) * g = c(f * g) = f * (cg)$ for $c \in \mathbb{C}$.
(iii) $f * g = g * f$.
(iv) $(f * g) * h = f * (g * h)$.
(v) $f * g$ is continuous.
(vi) $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$.

Lemma 3.2

Let f be integrable on the circle with $|f| \le B$. There are continuous functions f_k on the circle with $\sup|f_k| \le B$ and $\int_{-\pi}^{\pi} |f(x) - f_k(x)|\,dx \to 0$.

Proof. Assume f real (otherwise treat real and imaginary parts). Given $\epsilon > 0$ take a partition $-\pi = x_0 < \cdots < x_N = \pi$ with $U(P,f) - L(P,f) < \epsilon$. Let $f^*(x) = \sup_{x_{j-1} \le y \le x_j} f(y)$ on $[x_{j-1}, x_j)$; then $f^* \ge f$ and $\int (f^* - f) \le \int f^* - \int_L = U - L < \epsilon$. Now modify the step function f^* near each x_j: on $[x_j - \delta, x_j + \delta]$ replace it by the linear segment joining $f^*(x_j - \delta)$ to $f^*(x_j + \delta)$ (and match periodically at $\pm\pi$). The resulting $\tilde{f}$ is continuous, $|\tilde{f}| \le B$, and differs from f^* on a set of total length $\le 2N\delta$, where $|\tilde{f} - f^*| \le 2B$. With δ small, $\int |f^* - \tilde{f}| < \epsilon$. Hence $\int |f - \tilde{f}| < 2\epsilon$; take $\epsilon = 1/k$. ∎

Proof of Proposition 3.1.

Step 1 (continuous f, g). (i), (ii) are linearity of the integral. For (iii), substitute $y \mapsto x - y$ and use periodicity. For (iv), interchange the order of integration (continuous integrand on a square) and substitute. For (v): f is uniformly continuous, so $|f * g(x_1) - f * g(x_2)| \le \sup_y |f(x_1 - y) - f(x_2 - y)| \cdot \sup|g| \to 0$ (using (iii)). For (vi):

$$\widehat{f * g}(n) = \frac{1}{2\pi} \int \Big(\frac{1}{2\pi} \int f(y) g(x-y)\,dy \Big) e^{-inx}\,dx$$
$$= \frac{1}{2\pi} \int f(y) e^{-iny} \Big(\frac{1}{2\pi} \int g(x-y) e^{-in(x-y)}\,dx \Big)\,dy = \hat{f}(n)\hat{g}(n).$$

Step 2 (approximation). For integrable f, g, take continuous $f_k \to f$, $g_k \to g$ as in Lemma 3.2 (bounded by B). Then $f * g - f_k * g_k = (f - f_k) * g + f_k * (g - g_k)$ and $\sup_x |(f - f_k) * g(x)| \le \frac{B}{2\pi} \int |f - f_k| \to 0$, similarly for the second term. So $f_k * g_k \to f * g$ uniformly, which gives (v) (uniform limit of continuous functions) and (iii), (iv) by passing to the limit. For (vi), $\hat{f}_k(n) \to \hat{f}(n)$ since $|\hat{f}(n) - \hat{f}_k(n)| \le \frac{1}{2\pi} \int |f - f_k|$, and $\widehat{f_k * g_k}(n) \to \widehat{f * g}(n)$ by uniform convergence. ∎

2.4 Good kernels

Motivation. A family of kernels with total integral 1, bounded L^1 norms and mass concentrating at 0 reproduces f through convolution. The Dirichlet kernel fails the middle condition, which is why the partial sums need not converge, while the Fejér and Poisson kernels are good.

Definition (Good kernels)

A family $\{K_n\}$ of integrable 2π-periodic functions is a family of **good kernels** if (i) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x)\,dx = 1$ for all n;
(ii) $\int_{-\pi}^{\pi} |K_n(x)|\,dx \le M$ for all n;
(iii) for every $\delta > 0$, $\int_{\delta \le |x| \le \pi} |K_n(x)|\,dx \to 0$ as $n \to \infty$. (If $K_n \ge 0$, (ii) follows from (i).)
The same definition is used for families indexed by a continuous parameter.

Theorem 4.1

If $\{K_n\}$ is a family of good kernels and f is integrable on the circle, then $(f * K_n)(x) \to f(x)$ at every point of continuity x of f. If f is continuous, the convergence is uniform.

Proof. Let f be continuous at x and $\epsilon > 0$; choose δ with $|f(x - y) - f(x)| < \epsilon$ for $|y| < \delta$. By (i) and (iii),

$$(f * K_n)(x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y)[f(x - y) - f(x)]\,dy,$$

so, with $|f| \le B$,

$$|(f*K_n)(x)-f(x)| \le \frac{1}{2\pi}\int_{|y|<\delta}|K_n||f(x-y)-f(x)| + \frac{1}{2\pi}\int_{\delta\le|y|\le\pi}|K_n||f(x-y)-f(x)|$$
$$\le \frac{\epsilon M}{2\pi} + \frac{2B}{2\pi}\int_{\delta\le|y|\le\pi}|K_n(y)|dy,$$

using (ii), and the last term is $< \epsilon$ for large n by (iii). If f is continuous, it is uniformly continuous and δ is independent of x. ■

Since $S_N(f) = f * D_N$, one might hope $\{D_N\}$ is good. It satisfies (i), but not (ii): [Supplement] $|\sin((N+\frac{1}{2})\theta)|/|\sin(\theta/2)| \ge \frac{2}{|\theta|}|\sin((N+\frac{1}{2})\theta)|$, and integrating over $[-\pi,\pi]$ gives $\int|D_N| \ge c\log N$ (sum over the $2N$ subintervals where $|\sin((N+\frac{1}{2})\theta)| \ge \frac{1}{2}$). This is the root of the difficulties of pointwise convergence.

2.5 Cesàro and Abel summability: applications to Fourier series

Motivation. Averaging the partial sums (Cesàro) or damping the terms by $r^{|n|}$ (Abel) replaces D_N by the Fejér or Poisson kernel, which are good. Hence continuous functions are uniform limits of trigonometric polynomials, and the Poisson integral solves Dirichlet's problem in the disc.

2.5.1 Cesàro means and summation

For $s_n = \sum_{k=0}^{n} c_k$, the N-th **Cesàro mean** is $\sigma_N = \frac{s_0+\cdots+s_{N-1}}{N}$. $\sum c_k$ is **Cesàro summable** to σ if $\sigma_N \to \sigma$. Convergent series are Cesàro summable to their sum; $1-1+1-\ldots$ is Cesàro summable to $\frac{1}{2}$ but diverges.

2.5.2 Fejér's theorem

$\sigma_N(f)(x) = \frac{S_0(f)(x)+\cdots+S_{N-1}(f)(x)}{N} = (f*F_N)(x)$, where $F_N = \frac{D_0+\cdots+D_{N-1}}{N}$ is the **Fejér kernel**.

Lemma 5.1

$F_N(x) = \frac{1}{N}\frac{\sin^2(Nx/2)}{\sin^2(x/2)}$, and $\{F_N\}$ is a family of good kernels.

Proof. With $\omega = e^{ix}$, $D_n = \frac{\omega^{-n}-\omega^{n+1}}{1-\omega}$, so

$$NF_N = \frac{1}{1-\omega}\Big(\sum_{n=0}^{N-1}\omega^{-n} - \omega\sum_{n=0}^{N-1}\omega^{n}\Big) = \frac{1}{1-\omega}\Big(\frac{1-\omega^{-N}}{1-\omega^{-1}} - \frac{\omega(1-\omega^N)}{1-\omega}\Big) = \frac{\omega^{-N}-2+\omega^N}{(1-\omega)(1-\omega^{-1})},$$

and $\omega^{-N}-2+\omega^N = -(2\sin(Nx/2))^2$, $(1-\omega)(1-\omega^{-1}) = 2-2\cos x = (2\sin(x/2))^2$ with the signs matching ($NF_N \ge 0$). Good kernel: $F_N \ge 0$; $\frac{1}{2\pi}\int F_N = 1$ since each D_n has integral 1 (average of 1's); for $\delta \le |x| \le \pi$, $\sin^2(x/2) \ge c_\delta > 0$, so $F_N(x) \le \frac{1}{Nc_\delta}$ and $\int_{\delta\le|x|\le\pi} F_N \to 0$. ■

Theorem 5.2 (Fejér)

If f is integrable on the circle, its Fourier series is Cesàro summable to f at every point of continuity. If f is continuous, the Fourier series is uniformly Cesàro summable to f.

Proof. $\sigma_N(f) = f * F_N$ with $\{F_N\}$ good (Lemma 5.1); apply Theorem 4.1. ■

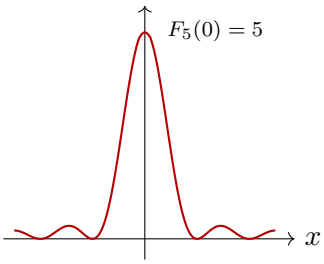


Figure 2: The Fejér kernel F_5: non-negative, integral 1, concentrating at 0.

Corollary 5.3

If f is integrable on the circle and $\hat{f}(n) = 0$ for all n, then $f = 0$ at all points of continuity of f.

Proof. All partial sums, hence all Cesàro means, vanish; apply Theorem 5.2. ■

Corollary 5.4

Continuous functions on the circle can be uniformly approximated by trigonometric polynomials.

Proof. $\sigma_N(f)$ is a trigonometric polynomial, and $\sigma_N(f) \to f$ uniformly (Theorem 5.2). ■

2.5.3 Abel means and summation

$\sum c_k$ is **Abel summable** to s if $A(r) = \sum c_k r^k$ converges for $0 \le r < 1$ and $A(r) \to s$ as $r \to 1^-$. Cesàro summable series are Abel summable to the same limit (not conversely: $\sum (-1)^k (k+1)$ is Abel summable to $\frac{1}{4}$ but not Cesàro summable). For $f \sim \sum a_n e^{in\theta}$, the **Abel means** are $A_r(f)(\theta) = \sum r^{|n|} a_n e^{in\theta} = (f * P_r)(\theta)$, by interchanging sum and integral (uniform convergence in the integration variable, $|a_n| \le \sup |f|$).

2.5.4 The Poisson kernel and Dirichlet's problem in the unit disc

Lemma 5.5

If $0 \le r < 1$, then $P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta+r^2}$, and $\{P_r\}_{0 \le r < 1}$ is a family of good kernels as $r \to 1^-$.

Proof. With $\omega = re^{i\theta}$, $P_r = \sum_{n \ge 0} \omega^n + \sum_{n \ge 1} \bar{\omega}^n = \frac{1}{1-\omega} + \frac{\bar{\omega}}{1-\bar{\omega}} = \frac{1-|\omega|^2}{|1-\omega|^2} = \frac{1-r^2}{1-2r\cos\theta+r^2}$. Good kernel: $P_r > 0$; $\frac{1}{2\pi}\int P_r = 1$ by term-by-term integration of the uniformly convergent series. For $\frac{1}{2} \le r < 1$ and $\delta \le |\theta| \le \pi$, $1 - 2r\cos\theta + r^2 = (1-r)^2 + 2r(1-\cos\theta) \ge c_\delta > 0$, so $P_r(\theta) \le \frac{1-r^2}{c_\delta} \to 0$. ■

Theorem 5.6

The Fourier series of an integrable function on the circle is Abel summable to f at every point of continuity. If f is continuous, it is uniformly Abel summable to f.

Proof. $A_r(f) = f * P_r$ with $\{P_r\}$ good; apply Theorem 4.1. ■

Theorem 5.7

Let f be integrable on the unit circle and $u(r,\theta) = (f * P_r)(\theta)$ for $0 \le r < 1$. Then:
(i) u is C^2 in the unit disc and $\triangle u = 0$.
(ii) If θ is a point of continuity of f, then $u(r,\theta) \to f(\theta)$ as $r \to 1$. If f is continuous, $u(r,\theta) \to f(\theta)$ uniformly as $r \to 1$.
(iii) If f is continuous, then u is the unique solution of $\triangle u = 0$ in the disc that satisfies (ii).

Proof.

(i) $u(r,\theta) = \sum a_n r^{|n|} e^{in\theta}$ with $|a_n| \le B$. For $r \le \rho < 1$ the series and the series of its first and second derivatives in r and θ (with factors $|n|^2$) converge uniformly, so u is C^2 (indeed C^∞) and can be differentiated termwise; each term $r^{|n|}e^{in\theta}$ is harmonic by the polar Laplacian of §1.2.2.

(ii) Theorem 5.6.

(iii) [Supplement] (The book leaves this to an exercise.) Let v be another solution and $w = u - v$: w is harmonic in the disc and $w(r,\theta) \to 0$ uniformly as $r \to 1$ (uniform in (ii) for both). Extending w by 0 on the circle gives a continuous function on the closed disc. *Maximum principle:* for $\epsilon > 0$, $w_\epsilon = w + \epsilon(x^2 + y^2)$ has $\triangle w_\epsilon = 4\epsilon > 0$, so w_ϵ has no interior maximum (there $\triangle w_\epsilon \le 0$). Hence $\max w_\epsilon$ is attained on the circle, where $w_\epsilon = \epsilon$. Letting $\epsilon \to 0$, $w \le 0$; applying this to $-w$, $w = 0$ (for complex w, take real and imaginary parts). ■